

Wavelength-Polarization Domain Synthetic Aperture Combining of Noncoherent Multi-beams for Turbulence-Resilient Satellite-to-Ground Laser Downlinks

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Abstract We propose a wavelength-polarization domain synthetic aperture combining scheme for turbulence-resilient satellite-to-ground downlinks based on VCM-FSM-enabled noncoherent multibeam transmission. This architecture supports all-weather uninterrupted communication under atmospheric turbulence and demonstrates 2.24 Tbit/s transmission over a 3000 km FSO equivalent link. ©2026 The Author(s).

Introduction

The space-air-ground integrated information network (SAGIN) has attracted significant attention for future 6G architecture, due to its broad coverage, ultra-high data rates, and strong interference resilience [1-3]. An increasing number of organizations—including ESA, JAXA, One Web, NASA, and SpaceX—are developing lightweight, cost-efficient, and high-capacity satellite communication terminals for deployment across multiple orbital regimes [4]. Low Earth Orbit (LEO) satellites, operating at altitudes of 300–3000 km, have become crucial elements of space-ground communication networks due to their rapid deployment, low latency, and operational robustness [5].

However, satellite-to-earth communication links still face challenges posed by atmospheric turbulence. Atmospheric turbulence can cause intensity scintillation of the laser beam, distort the wavefront, and increase beam pointing errors, leading to a reduced SNR and exacerbated signal distortion at the receiver [6]. To mitigate atmospheric turbulence, researchers have conducted a series of studies. A single-input single-

output (SISO) communication scheme is highly dependent on system stability and is vulnerable to signal interruption under atmospheric turbulence, thereby requiring higher link power margins and more precise receiving and tracking systems. In contrast, diversity techniques, such as polarization diversity and spatial diversity, offer a concise and efficient approach [7-9]. However, the potential of jointly exploiting wavelength and polarization dimensions for turbulence-resilient multibeam combining in satellite-to-ground links remains largely unexplored.

In this paper, we propose a turbulence-resilient laser communication system using noncoherent multibeam transmission and wavelength-polarization domain synthetic aperture combining. The voice-coil-motor-driven fast-steering-mirror (VCM-FSM) array enables multibeam downlink pointing and overlap. A global fusion-enhanced multi-branch network (GFE-MBN) is developed for adaptive combining of wavelength-polarization domain signals. Compared with the SISO link, the proposed synthetic aperture combining scheme achieves an 8.4 dB Q-factor gain at the

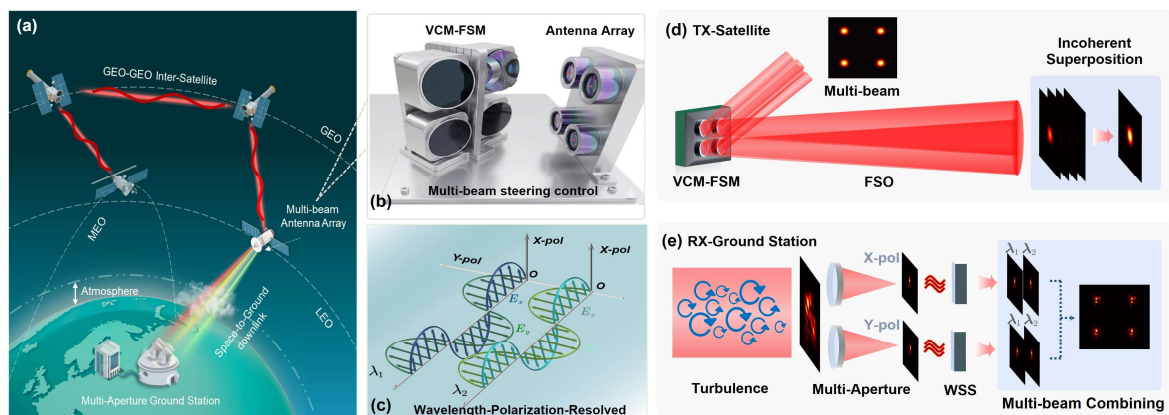


Fig.1: (a) Space-air-ground integrated information network. (b) Multi-beams steering control. (c) Wavelength-polarization resolved noncoherent beam (d) Incoherent superposition from the satellite. (e) Synthetic aperture combining from the ground station.

3.8e-3 FEC threshold, while maintaining above-threshold Q-factor under turbulence conditions $r_0 = 2mm$ and $r_0 = 1.5mm$. The results show the scheme effectively mitigates turbulence-induced communication interruption. In addition, this architecture demonstrates 2.24 Tbit/s data transmission over a 3000 km equivalent FSO link.

Principle

An VCM-FSM array enables beam scanning and multi-directional pointing, providing wide angular coverage, compact form factor, and high pointing accuracy, as illustrated in Fig. 1b. In the proposed scheme, the satellite communication payload transmits four noncoherent beams corresponding to two wavelengths and two orthogonal polarizations, denoted as $\lambda_1^X, \lambda_1^Y, \lambda_2^X$ and λ_2^Y . A 2×2 VCM-FSM array is used to realize far-field beam superposition at the required pointing angle for the satellite-to-ground link. At the ground station, a dual-aperture receiver first separates the turbulence-affected beams by polarization and then separates the wavelength channels, enabling two-dimensional wavelength-polarization resolution. This enables crosstalk-free diversity reception for multibeam downlinks, as shown in Fig. 1(d) and 1(e).

A global fusion-enhanced multi-branch neural network (GFE-MBN) is employed to combine multiple received signals at the ground data processing module, as shown in Fig. 2(c). The network adopts an $N+1$ branch architecture. The first N branches independently extract features from the constellation points of the N received signals, whereas the $(N+1)$ th branch takes their

weighted fusion as input. The weighted-sum branch is generated by applying a two-dimensional (2D) convolutional kernel to the concatenated outputs of the first N branches:

$$\text{Conv2D}(\text{Concat}(X_1, X_2, \dots, X_N), [\omega_1, \omega_2, \dots, \omega_N]) \quad (1)$$

$$= \omega_1 X_1 + \omega_2 X_2 + \dots + \omega_N X_N$$

Each of the $N+1$ branches is fed into a subnet containing one hidden layer ($h_1, h_2, \dots$, and h_{n+1}). The outputs of the subnets are concatenated (h_{cat}) and then passed through another hidden layer (h_z) to produce the final output:

$$h_{cat} = [\sigma(h_1); \sigma(h_2); \dots; \sigma(h_{n+1})], \quad (2)$$

$$h_1, h_2, \dots, h_{n+1}, h_z \in R^{M \times 1}, h_{cat} \in R^{(n+1)M \times 1}$$

where $\sigma(\cdot)$ denotes the activation function. Since the constellation points are complex-valued, a complex-valued neural network is employed. This branch structure allows the network to learn both the individual features of each signal branch and the correlations among different branches, thus improving signal fusion performance.

Experimental setup

We construct a noncoherent four beam transmission system for atmospheric turbulence based on wavelength-polarization domain synthetic aperture combining, as shown in Fig. 2 (a). The system employs master-slave frequency combs to generate 40 four-beam configurations, achieving 2.24Tbit/s data transmission over a 3000 km equivalent link. Each master-slave wavelength pair forms noncoherent four beams under two orthogonal polarizations, namely $\lambda_m^X, \lambda_m^Y, \lambda_s^X$ and λ_s^Y , as shown in Fig. 2(b). An amplified spontaneous emission (ASE) source was

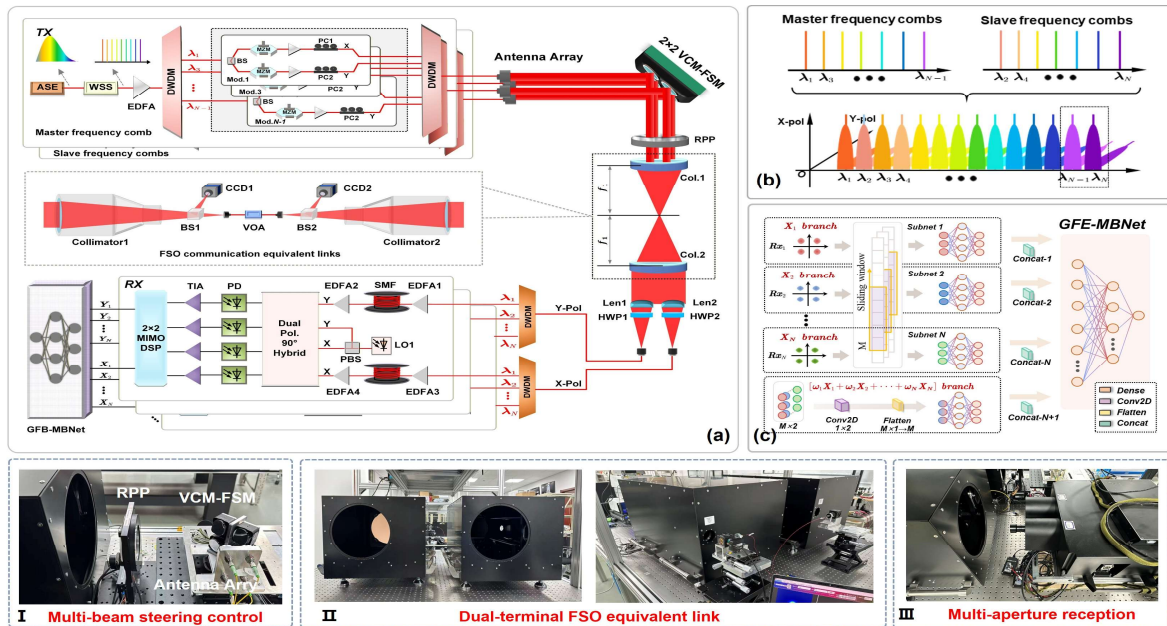


Fig.2: (a) Noncoherent four beam transmission system for atmospheric turbulence based on wavelength-polarization domain synthetic aperture combining. (b) Master-slave frequency combs with polarization diversity. (c) Global fusion-enhanced multi-branch network.

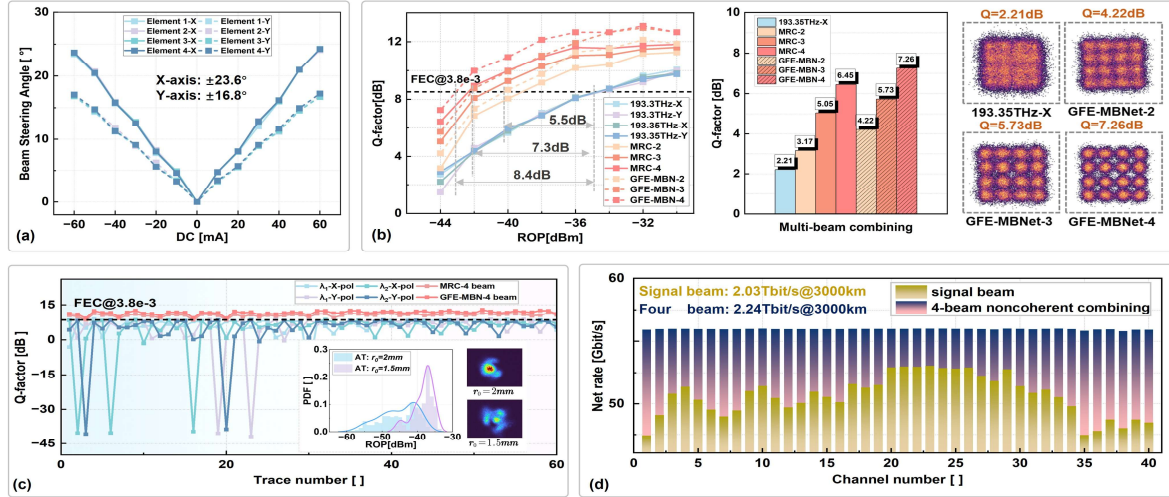


Fig.3: (a) Beam steering Angle of the VCM-FSM. (b) Results of four-beam 16QAM transmission based on wavelength-polarization domain synthetic aperture combining. (c) Q-factor for 16QAM under two different turbulence intensities. (d) Net data rate of multibeam 4QAM transmission based on master-slave frequency combs.

spectrally shaped using a wavelength selective switch (WSS) to generate a 40-line master frequency comb with 100-GHz spacing. After DWDM demultiplexing, each spectral line was split by a beam splitter (BS) and then modulated using QAM at a symbol rate of 28 GHz. PCs were used to adjust the two output branches to λ_m^x and λ_m^y , respectively. Similarly, the slave comb also generates λ_s^x and λ_s^y , which together with the master comb form four beams. The multiplexed four beams are transmitted through 2×2 antenna array, and VCM-FSM array is used to steer the beams into the equivalent FSO link. The rotating phase plate (RPP) introduced wave-front distortions emulating atmospheric turbulence. At the receiver, the HWP in the dual-aperture antenna separates the orthogonal polarizations. The WSS then filters the master and slave wavelengths, enabling crosstalk-free diversity reception of the four beams. The optical signals are fed into a digital coherent receiver for optical-to-electrical conversion, and the GFE-MBN is finally used to achieve optimal combining of the four received signals.

Results

Figure 3(a) shows the beam steering ranges of the VCM-FSM along the X and Y axes, which are $\pm 23.6^\circ$ and $\pm 16.8^\circ$, respectively. The designed mirror has a 40 mm aperture and a pointing accuracy better than $30 \mu\text{rad}$, meeting the requirements of satellite-to-ground downlinks. Figure 3(b) shows the results of four-beam 16QAM transmission based on wavelength-polarization domain combining. Compared with the SISO, the GFE-MBN-based two-beam, three-beam, and four-beam combining schemes provide Q-factor gains of 5.5 dB, 7.3 dB, and 8.4 dB, respectively, at a BER threshold of 3.8×10^{-3} . Notably, GFE-MBN

achieves a higher combining gain than conventional maximum ratio combining (MRC). This is attributed to its ability to learn nonlinear noise of system and exploit inter-signal correlations through the $(N+1)$ th branch. We designed a bidirectional FSO equivalent link to emulate indoor FSO far-field transmission based on the Airy disk far-field power density expression [10]. Figure 3(c) shows the Q-factor of the four-beam 16QAM signal under different turbulence conditions over a 500 km equivalent link, measured 60 times at 2 s intervals. At turbulence strengths of $r_0 = 2\text{ mm}$ and 1.5 mm , the insets show the corresponding ROP distributions, where the four-beam combined signal remains above the BER threshold compared with the single-beam. Figure 5 shows the results of 40 four-beam transmission groups generated by master-slave frequency combs. The system achieves 2.24 Tbit/s over a 3000 km equivalent link, with a 180 Gbit/s capacity gain. The proposed architecture significantly improves both turbulence resilience and transmission capacity, highlighting its strong potential for satellite communication applications.

Conclusions

In this paper, we propose a turbulence-resilient laser communication system using noncoherent multibeam transmission and wavelength-polarization domain synthetic aperture combining. The results show four-beam combining achieves an 8.4 dB Q-factor gain and remains above the BER threshold under different turbulence conditions. In addition, the system achieves 2.24 Tbit/s over a 3000 km equivalent link, with a 210 Gbit/s capacity gain.

Acknowledgements

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